

Efficient State Space Models for Body-Conducted Speech Enhancement

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Abstract

Body-conducted microphones offer noise-robust speech capture but produce signals with bandwidth limitations and spectral distortions. We investigate Structured State Space Models (SSMs) for enhancing soft in-ear microphone speech from the VibravOx dataset. Our experiments demonstrate that S5 (597K parameters) achieves comparable performance to Mamba (4.1M parameters) with $7\times$ fewer parameters on clean speech, and *outperforms* Mamba on downstream ASR (WER). Crucially, S5 converges robustly under joint clean+noisy training, while Mamba requires a two-stage curriculum (pretraining then fine-tuning) to achieve competitive performance. This demonstrates that fixed-parameter SSMs offer superior training stability for body-conducted speech enhancement.

Index Terms: speech enhancement, state space models, body-conducted speech, S5, Mamba

1. Introduction

Voice interfaces have become ubiquitous, yet their reliability degrades significantly in real-world conditions. Word error rates increase from 3% in quiet environments to over 30% in noisy settings, limiting practical utility [1]. Beyond noise, current voice assistants struggle with speech impairments such as dysarthria, excluding users who could benefit most from hands-free interaction. Privacy concerns further complicate adoption: accidental activations, unintended recordings, and data misuse have led to legal settlements and eroded user trust.

Body-conducted microphones offer a compelling solution to these challenges. By capturing speech through bone and tissue vibration rather than air conduction, in-ear devices naturally attenuate ambient noise while enabling private, subvocal communication [2]. However, these signals suffer from bandwidth limitations and spectral distortions that degrade speech quality and intelligibility. The VibravOx dataset [2, 3] provides a benchmark for this challenging enhancement task, with EBEN [4] establishing the current state-of-the-art using a GAN-based architecture.

Recent advances in Structured State Space Models (SSMs) offer a compelling alternative to attention-based architectures for sequence modeling. SSMs achieve linear time complexity while effectively capturing long-range dependencies [5]. The S5 architecture [6] simplifies the original S4 formulation, while Mamba [7] introduces selective state spaces for adaptive modeling. These models have shown promise for speech enhancement [8, 9] and audio processing [10], but their application to body-conducted speech remains unexplored.

In this work, we investigate SSM-based architectures for soft in-ear microphone speech enhancement. Our key contributions are:

- We present the first application of Structured State Space Models to body-conducted speech enhancement
- We demonstrate that S5 (597K parameters) achieves comparable performance to Mamba (4.1M parameters) with $7\times$ **fewer parameters**, and generalizes better to noisy conditions with improved downstream ASR
- We show fixed-parameter SSMs provide a stable, streaming-compatible alternative to specialized baselines, improving WER on body-conducted speech
- We provide comprehensive evaluation using intrusive (PESQ, ESTOI, SI-SDR), non-intrusive (NISQA), and downstream (WER) metrics

2. Related Work

Body-Conducted Speech Enhancement. Body-conducted microphones capture speech through bone and tissue vibration, offering noise robustness but suffering from bandwidth limitations due to low-pass filtering by biological pathways [2]. The VibravOx dataset provides paired recordings from five body-conducted sensors and an airborne reference, enabling supervised enhancement. EBEN [4] addresses bandwidth extension using a GAN with pseudo-quadrature mirror filter (PQMF) decomposition, achieving state-of-the-art perceptual quality. TRAMBA [11] combines Transformer and Mamba for bone conduction speech on mobile platforms, but requires 23M parameters.

State Space Models. Structured State Space Models (SSMs) provide an efficient alternative to attention for sequence modeling, achieving linear complexity in sequence length [5]. The original S4 model [5] introduced HiPPO-initialized state matrices for long-range dependencies. S5 [6] simplifies this with diagonal complex-valued states and parallel associative scans, enabling efficient training. Mamba [7] introduces *selective* state spaces where transition parameters are input-dependent, trading fixed dynamics for adaptive modeling at increased parameter cost.

SSMs for Speech. SSMs have shown promise across speech tasks. SEMamba [8] applies Mamba to speech enhancement on VoiceBank-DEMAND, while aTENNuate [9] targets hearing aid applications. Speech Slytherin [12] provides comprehensive analysis across ASR, TTS, and enhancement. However, SSM application to body-conducted speech—where bandwidth extension rather than denoising is primary—remains unexplored. We address this gap by comparing fixed-parameter (S5) and selective (Mamba) SSMs on the VibravOx benchmark.

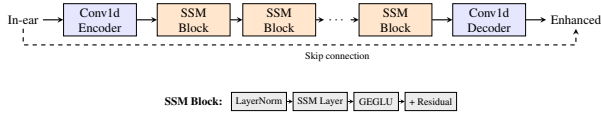


Figure 1: *Model architecture. Both S5 and Mamba share this encoder-decoder structure with 6 stacked SSM blocks. Each SSM block contains LayerNorm, the SSM layer (S5 or Mamba), GEGLU feedforward, and residual connection. A skip connection preserves low-frequency content.*

3. Method

We compare two SSM architectures that represent different design philosophies: S5 with fixed state dynamics and Mamba with input-dependent selection. Both models share an encoder-decoder structure (Figure 1): a Conv1d frontend projects input to 128 dimensions, followed by 6 stacked SSM blocks (each containing LayerNorm, SSM layer, GEGLU feedforward, and residual connection), and a Conv1d output head. An additive skip connection from input to output preserves low-frequency content. All convolutions use causal padding, and SSMs are unidirectional, enabling streaming inference.

S5 [6] uses diagonal complex-valued state spaces initialized with HiPPO [5]. Crucially, the state transition matrices are *fixed learned parameters*, enabling efficient parallel scans during training. Each block combines an SSM layer with GEGLU feedforward, totaling **597K parameters**.

Mamba [7] introduces *selective* state spaces where the matrices B , C , and discretization step Δ are recomputed from the input at each timestep. This input-dependent mechanism enables adaptive memory allocation but requires sequential computation and increases model size to **4.1M parameters** ($7\times$ larger).

3.1. Loss Functions

The choice of loss function significantly impacts enhancement quality. We investigate three configurations: (1) SI-SDR combined with multi-resolution STFT loss ($\mathcal{L} = -\text{SI-SDR}/10 + \mathcal{L}_{\text{MRSTFT}}$), (2) SI-SDR alone, and (3) L1 reconstruction loss. The MRSTFT loss [13] captures spectral structure at multiple time scales, complementing the time-domain SI-SDR objective.

3.2. Training

Models are trained on VibravOx *speech_clean* (20,981 samples) at 16 kHz with 1-second segments. We use AdamW optimization with learning rates of 10^{-3} for S5 and 3×10^{-4} for Mamba, ReduceLROnPlateau scheduling, gradient clipping at 1.0, and early stopping with patience 10.

4. Experiments

4.1. Setup

We evaluate on VibravOx [2], a French speech corpus recorded from 188 participants (97F/91M) with paired recordings from five body-conducted sensors and an air-conducted headset reference. We focus on the soft silicone in-ear microphone, which captures speech through ear canal tissue vibration. The corpus includes approximately 45 hours of audio per sensor, including 35 hours of speech (*speech_clean*) and noisy conditions (*speech_noisy*) with real environmental noise. We use two test sets: E1 (*speech_clean*, 3,064 samples) for matched

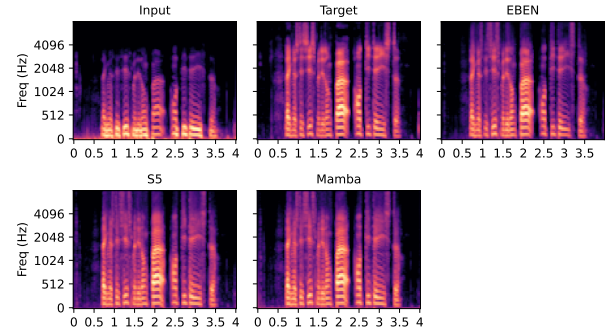


Figure 2: *Mel spectrograms on a speech_clean sample. The in-ear input shows bandwidth limitations; S5 and Mamba recover high-frequency structure similar to the headset target.*

conditions and E2 (*speech_noisy*, 175 samples) for noise robustness evaluation. E2 is small, so results should be interpreted as indicative.

For SSMs, we compare two training regimes: (C) training on *speech_clean* only, and noise adaptation via either (C+N) joint training on both splits or (C→N) fine-tuning a clean-pretrained model on *speech_noisy* with $\text{lr}=10^{-4}$ and fresh optimizer state. We include three baselines: unprocessed input (Identity), EBEN [4] (GAN-based, pretrained on VibravOx), and MetricGAN+ [14] (pretrained on VoiceBank-DEMAND, zero-shot transfer).

We report intrusive metrics, PESQ [15], ESTOI [16], SI-SDR [17], alongside non-intrusive quality (NISQA [18]) and downstream ASR performance (WER via Whisper-base [1]). WER can exceed 100% because it is computed as $(S + D + I)/N$, where insertion errors can make the numerator exceed the reference word count N . The in-ear signal is severely band-limited and spectrally distorted, making ASR challenging even after enhancement. Note that Identity SI-SDR is highly negative because the body-conducted input and air-conducted target are physically distinct signals with inherent phase and magnitude differences, not merely additive noise.

4.2. Results

Because our target application is speech understanding, we treat downstream ASR (WER) as the primary metric; perceptual metrics are secondary.

Table 1 summarizes performance across both test conditions. On clean speech (E1), EBEN achieves the highest perceptual quality (PESQ 2.07, NISQA 2.92), while SSMs excel at time-domain reconstruction (SI-SDR 2.52 for Mamba). On noisy speech (E2), noise adaptation improves all architectures: S5 with joint training (C+N) achieves the lowest WER (116%) and competitive SI-SDR (-1.22 dB), while EBEN C→N achieves the best perceptual scores (PESQ 1.53, NISQA 2.19). Notably, joint C+N training proved highly unstable for Mamba, failing to converge within our training budget (72-hour SLURM limit); we therefore report Mamba with fine-tuning C→N (-1.02 dB SI-SDR). Figure 2 shows representative spectrograms comparing enhancement outputs.

4.3. Analysis

On clean speech (E1), both SSM architectures perform comparably, with Mamba achieving slightly higher PESQ (1.84 vs

Table 1: Results on E1 (speech_clean, n=3064) and E2 (speech_noisy, n=175). Best per-column in **bold**. C+N = joint training, C→N = fine-tuned.

Model	Train	Params	E1: Clean Speech					E2: Noisy Speech				
			PESQ	ESTOI	SI-SDR	NISQA	WER(%)↓	PESQ	ESTOI	SI-SDR	NISQA	WER(%)↓
Identity	—	—	1.45	0.69	-10.0	2.57	53	1.21	0.51	-11.8	1.80	144
EBEN	C	1.9M	2.07	0.80	0.14	2.92	38	1.35	0.52	-3.79	2.09	193
EBEN	C→N	1.9M	1.99	0.77	-0.42	2.62	58	1.53	0.54	-3.18	2.19	150
MetricGAN+	ext	1.9M	1.40	0.64	-9.69	2.27	76	1.22	0.42	-9.75	2.14	281
S5	C	597K	1.78	0.75	1.72	2.53	34	1.27	0.49	-4.30	1.64	175
S5	C+N	597K	1.80	0.75	1.93	2.44	61	1.48	0.53	-1.22	1.87	116
Mamba	C	4.1M	1.84	0.76	2.52	2.52	35	1.27	0.49	-3.55	1.67	175
Mamba	C→N	4.1M	1.80	0.73	1.70	2.35	66	1.43	0.51	-1.02	1.87	202

C=clean only, C+N=joint training on clean+noisy, C→N=pretrained on clean then fine-tuned on noisy, ext=external (VoiceBank-DEMAND pretrained). Mamba (C+N) omitted: did not converge within 72-hour training limit (7 epochs).

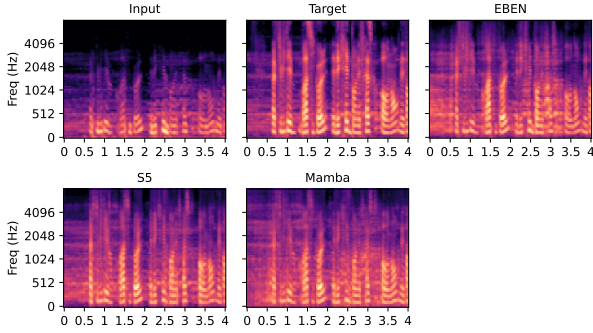


Figure 3: Mel spectrograms on a noisy sample. S5 (C+N) effectively removes background noise while preserving speech formants (1–4 kHz). Mamba (C→N) introduces subtle high-frequency artifacts above 6 kHz, which correlates with its degraded WER (202% vs 116%). EBEN achieves the cleanest output by perceptual metrics but with reduced high-frequency detail.

1.78). However, this comes at the cost of $7\times$ more parameters (4.1M vs 597K), making S5 substantially more parameter-efficient.

Noise adaptation improves E2 performance across all architectures, but the optimal strategy differs. S5 converges robustly with joint C+N training, achieving SI-SDR -1.22 dB and the lowest WER (116%). In contrast, we found joint training to be highly unstable for Mamba—despite sweeping learning rates (10^{-4} – 10^{-3}) with gradient clipping and early stopping, our C+N runs failed to converge within our 72-hour budget (7/100 epochs at ~ 9 h/epoch), with validation SI-SDR at -2.4 dB and still improving when terminated. To obtain a competitive Mamba model for noisy speech, we employed a two-stage curriculum: pretraining on clean speech followed by fine-tuning on noisy speech (C→N). This asymmetry reveals a fundamental difference in training stability: S5’s fixed state dynamics enable robust optimization on heterogeneous data, while Mamba’s input-dependent parameters require more constrained training curricula.

Notably, S5 achieves the lowest E2 WER (116%) with $7\times$ fewer parameters than Mamba. While Mamba C→N achieves slightly better SI-SDR (-1.02 vs -1.22 dB), S5’s superior WER suggests better preservation of speech intelligibility. For resource-constrained wearable deployment, this trade-

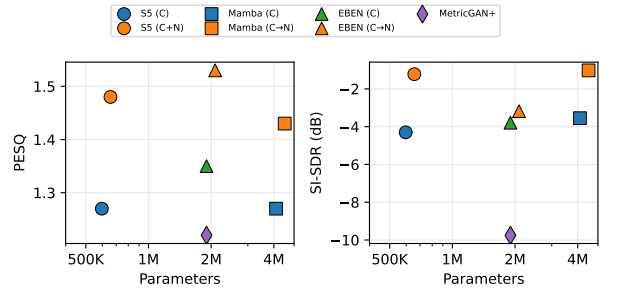


Figure 4: Parameters vs PESQ (left) and SI-SDR (right) on E2 (speech_noisy). Models exposed to noisy data (C+N or C→N) are indicated by orange markers with distinct black outlines. S5 (C+N) achieves competitive performance with $7\times$ fewer parameters than Mamba.

off strongly favors S5.

Among baselines, EBEN C→N achieves the best perceptual quality on E2 (PESQ 1.53, NISQA 2.19), demonstrating the effectiveness of GAN-based training for perceptual metrics. However, S5 achieves substantially better WER (116% vs 150%), indicating superior preservation of speech intelligibility for downstream ASR—our primary objective for wearable speech understanding. This metric-ASR mismatch is consistent with prior work showing perceptual scores do not always correlate with recognition accuracy [1]. MetricGAN+ fails to transfer from VoiceBank-DEMAND to body-conducted speech.

Loss Function Ablation. In ablation on S5 (C), SI-SDR-only loss achieved the highest SI-SDR (2.82 dB) while the composite SI-SDR+MRSTFT loss yielded the best perceptual quality (PESQ 1.78), confirming that multi-objective losses benefit body-conducted speech enhancement.

4.4. Computational Efficiency

To evaluate scalability, we measure inference speed on long audio segments. We construct these by concatenating consecutive E1 test samples from the same speaker, yielding 602 segments of ~ 30 s, 297 of ~ 60 s, and 142 of ~ 120 s. We report real-time factor (RTF = audio duration / processing time; higher is faster) in Table 2.

EBEN achieves the highest throughput (547 – $903\times$) due to its fully convolutional architecture, followed by MetricGAN+ (189 – $244\times$). Among SSMs, S5 is $1.5\times$ faster than Mamba

Table 2: Computational efficiency (RTF $\times$, higher=faster) on long segments. VRAM is peak memory using parallel scan; for deployment, recurrent mode requires <1GB.

Model	30s	60s	120s	VRAM
EBEN	547	830	903	0.4GB
MetricGAN+	189	203	244	0.1GB
S5	102	87	67	19.6GB
Mamba	68	64	52	16.6GB

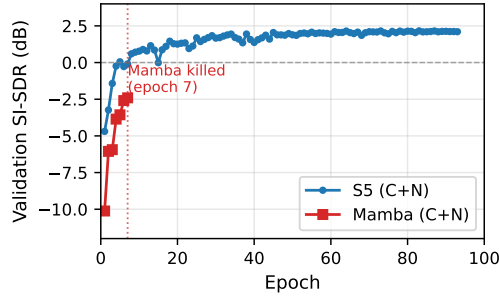


Figure 5: Validation SI-SDR during joint C+N training. S5 exhibits stable convergence to +2.1 dB, while Mamba fails to reach positive SI-SDR before the 72-hour compute budget is exhausted (epoch 7), necessitating the C $\rightarrow$ N curriculum in Table 1.

(102 $\times$ vs 68 $\times$ on 30s). Notably, SSM inference time scales super-linearly with sequence length due to state accumulation overhead, causing RTF to *decrease* on longer segments. In contrast, convolutional models scale sub-linearly, with RTF *increasing* as fixed overhead is amortized. The high VRAM usage (16–20GB) reflects our use of parallel scan inference for benchmarking; in recurrent mode, SSMs achieve O(1) memory complexity with respect to sequence length, enabling deployment on memory-constrained devices. However, EBEN’s GAN-based architecture requires adversarial training, adding optimization complexity not present in SSM training.

4.5. Discussion

The noise adaptation results reveal a fundamental difference in training stability between SSM architectures (Figure 5). S5’s fixed state transition matrices, initialized with HiPPO [5], provide a stable inductive bias that enables robust optimization on heterogeneous (clean+noisy) data. Joint C+N training exposes S5 to both conditions simultaneously, and its fixed dynamics learn robust representations efficiently.

In contrast, Mamba’s input-dependent parameters introduce optimization challenges. Because the state transitions are recomputed dynamically from each input, joint training on data with vastly different SNR distributions proved unstable—our Mamba C+N runs failed to converge within our training budget. Mamba required a constrained C $\rightarrow$ N fine-tuning curriculum to successfully adapt to noise, whereas S5 succeeded seamlessly with joint training. From a practitioner standpoint, this sensitivity is consequential: under common cluster time limits, Mamba’s joint-training regime may be impractical without substantial optimization or architectural changes.

Compared to EBEN—a domain-specialized model developed alongside VibravOx—S5 offers complementary advan-

tages: stable optimization under mixed-condition training, recurrent state updates enabling streaming inference, and improved downstream ASR despite lower perceptual scores. Including EBEN as a strong “home-field” baseline strengthens our evaluation; S5’s WER gains indicate SSMs generalize effectively to this modality.

We note that Mamba at a comparable parameter count to S5 (702K, d.state=16) failed across all 9 training attempts with negative SI-SDR, while S5 succeeded consistently at 597K parameters. This suggests a minimum capacity threshold for selective SSMs on this task, reinforcing S5’s advantage in parameter-constrained settings.

The speech_noisy fine-tuning set contains only 1,220 samples (~50 minutes of audio), which may limit adaptation for all models. With more noisy training data, the gap between S5 and Mamba could narrow further.

5. Conclusion

We presented the first application of Structured State Space Models to body-conducted speech enhancement. Our experiments on the VibravOx dataset reveal that S5 achieves superior noise robustness with 7 $\times$ fewer parameters than Mamba (597K vs 4.1M). Crucially, S5 converges robustly under joint clean+noisy training, while Mamba requires a two-stage curriculum to achieve competitive performance. S5’s combination of parameter efficiency, noise robustness, and training stability makes it well-suited for resource-constrained wearable deployment.

Our results have practical implications for accessibility and privacy. Body-conducted microphones enable robust speech capture in noisy environments and support private, subvocal communication, addressing key limitations of current voice interfaces. S5’s parameter efficiency and noise robustness make it suitable for resource-constrained wearable devices, potentially enabling always-on speech enhancement for users with hearing impairments or those working in challenging acoustic environments.

Limitations and Future Work. Our evaluation is limited to a single body-conducted sensor (soft in-ear microphone) and French speech. The noisy test set (E2) contains only 175 samples, limiting statistical power. The noisy fine-tuning set contains only 1,220 samples, which may limit adaptation for all models.

Future work will explore several directions: (1) *adversarial training* with a lightweight discriminator to improve perceptual quality (NISQA) while maintaining S5’s signal fidelity, EBEN’s high NISQA scores suggest GAN-based training could benefit SSMs; (2) *multi-band processing* to separately handle low-frequency content (well-preserved in body conduction) and high-frequency reconstruction; (3) *U-Net style skip connections* within the S5 stack for better gradient flow; (4) streaming inference for real-time applications; (5) cross-sensor generalization to other body-conducted modalities (throat, temple); and (6) evaluation on English and other languages.

Audio samples and code available at <https://anonymous.4open.science/r/ssm-body-speech-39C1/README.md>.

6. Acknowledgments

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